

operates as the master and one or more devices operate as slaves to send and receive data.

Bluetooth 5 is the latest version of this technology. It consumes low power and transfers data at 2Mbps speed. It employs Gaussian Frequency Shift Keying (GFSK) modulation technique over 2.4GHz to 2.483GHz spectrum.

It allows hands-free phone calls with a wireless cellular headset, speaker or car's audio system. It is widely used to establish a wireless connection between a mobile phone or computer and other electronic devices, including mouse, keyboard, printer, digital camera, speaker, etc.

Bluetooth 5 version offers two interfaces for the Internet of Things (IoT) devices: one for low energy operation to transmit fewer data over much longer distances, and the other for transmitting twice the data over a shorter range.

Bluetooth 5 allows users to play audio on two connected devices at the same time. People can send audio to multiple rooms in a house, create stereo effect in one space, or share audio between two sets of headphones.

Another area in which Bluetooth 5 adds improvement is Beacon technology; businesses such as retail can beam messages to nearby potential customers with offers or advertisements. People can opt out of this functionality by turning off location services and checking app permissions for retail stores.

Beacon technology can also facilitate indoor navigation, such as in an airport or shopping mall, and makes it easier for warehouses to track inventory.

Wi-Fi 6

A Wi-Fi setup consists of a wireless router that links many devices with an Internet connection depending on router configuration. Wi-Fi 6, also known as 802.11ax Wi-Fi, is the next-generation standard in Wi-Fi technology. It utilises 2.4GHz and 5GHz bands, and provides 9.6Gbps data transfer rate.

With its 1024-QAM (Quadrature amplitude modulation) technique, each symbol carries ten bits rather than eight bits, improving raw speeds by 25 per cent compared to 802.11ac 256-QAM. There are two key technologies that improve efficiency and speed up Wi-Fi 6 connections: multi-user

multiple input multiple outputs (MU-MIMO) and orthogonal frequency division multiple access (OFDMA).

If we assume a Wi-Fi connection as a series of delivery trucks delivering data packets to devices, with OFDMA each truck can deliver multiple parcels to multiple devices simultaneously, improving efficiency for both uploads and downloads. Its 8x8 MU-MIMO technique allows more than eight streams available for streaming, downloading, torrenting, playing virtual reality/augmented reality (VR/AR), massively multiplayer online games or role-playing games.

Target wake time feature allows devices to decide when and how often they will wake up to send or receive data, thus, increasing device sleep time and improving battery life of mobile and IoT devices. Base service station colour marks frames from neighbouring networks to ignore interference from the neighbours' wireless networks.

Wi-Fi 6 provides communication distance of about a hundred metres, making it suitable for quality connectivity in locations with hundreds or thousands of connected devices such as stadiums and other public venues, as well as corporate networks utilising time-sensitive, high-bandwidth and low-latency applications. Wi-Fi 6 devices operate efficiently in the most dense and dynamic connectivity settings like airports and railway stations. It provides better security features with the latest Wi-Fi Protected Access 3 (WPA3).

Near-field communication

NFC is often used for secure payment applications, fast passes, sharing information and wireless charging applications. It operates at 13.56MHz ISM frequency with a maximum range of around ten centimetres. This provides a more secure connection, which is usually encrypted. An inbuilt NFC tag allows phones, tablets, laptops and other devices to easily share data with other NFC-equipped devices.

NFC has evolved from radio-frequency identification (RFID) technology. It uses small, flat tags that can be attached to anything and used for identification, location, tracking and inventory management. When a reader unit is nearby, it transmits



NFC for payments (Credit: <https://tender-retail.acceo.com>)

a high-power RF signal to the tag and reads the data stored in the memory.

NFC can be used for mobile payments and as an alternative to credit cards. It also enables transferring data between two NFC-enabled devices. With NFC, a connection is automatically started when another NFC device enters the previously-specified range. Once in range, the two devices instantly communicate and send prompts to the user to initiate mobile payments, data sharing or wireless charging.

NFC wireless charging mode allows contactless transfer of up to 1W power to small devices like Bluetooth headset, fitness tracker or smartwatch, and other IoT devices.

Zigbee 3.0

Zigbee is a standard wireless communication protocol that defines physical and Media Access Control (MAC) layers to handle many devices at low data rates. It uses 802.15.4 transceiver as base and operates at 868MHz, 902MHz to 928MHz, and 2.4GHz frequencies. It can build large mesh networks for device monitoring and control, handling up to 65,000 nodes. It can also support multiple types of radio networks such as point-to-point and point-to-multipoint. It can transfer data wirelessly over a distance of up to hundred metres at 250kB/s speed.

Zigbee technology is best suited for low-latency and low-energy consumption at lower bandwidths. It can transfer data in two modes: beacon and non-beacon. When there is no data communication from end devices,